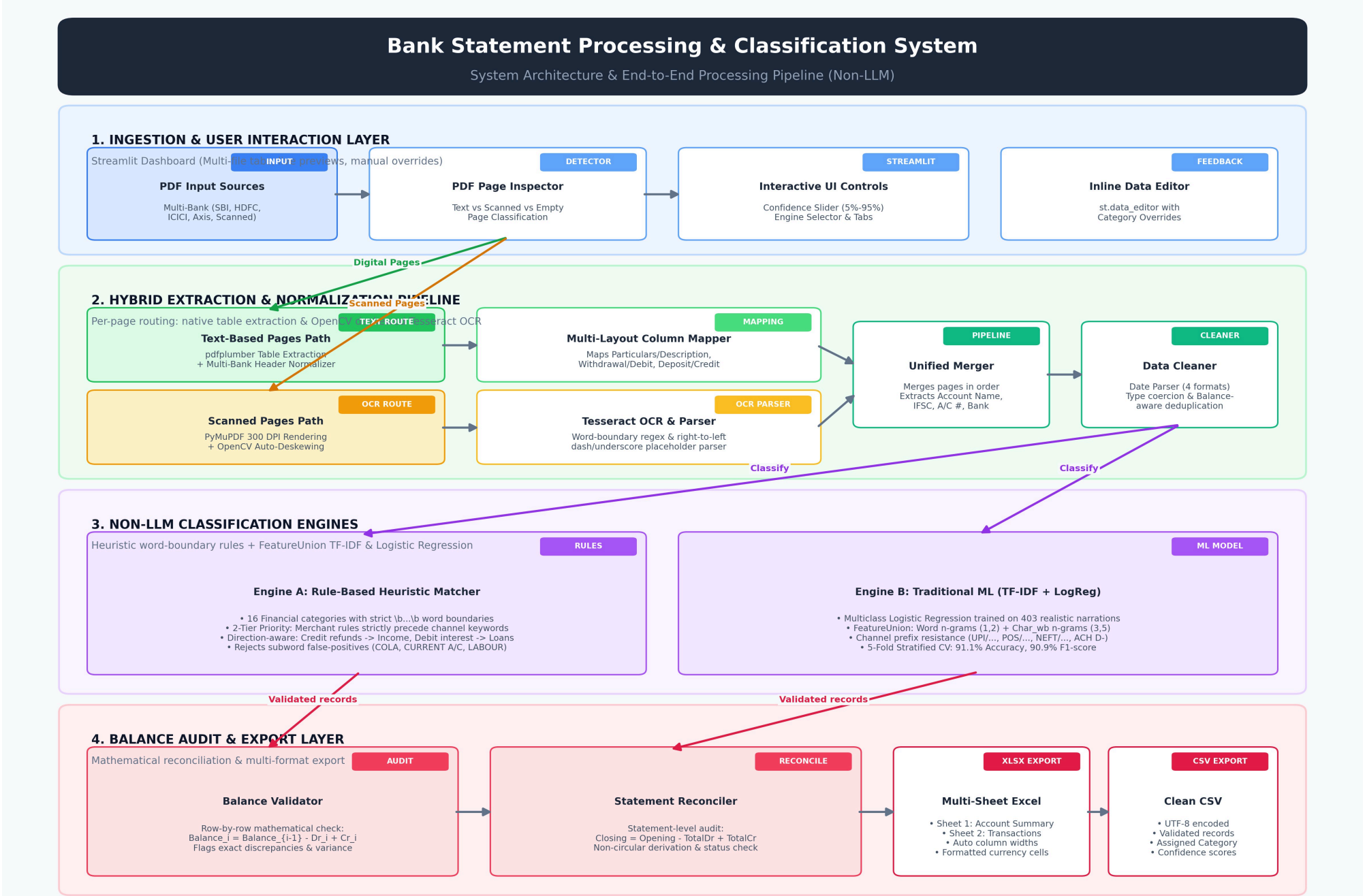
Bank Statement Processing & Classification System

System Architecture & Processing Pipeline (Non-LLM) | Befree AI & Automation Assessment



System Architecture Specifications

Component Responsibilities, Algorithms & Non-LLM Classification Logic

1. Presentation & Interaction Layer (Streamlit)

? Multi-file upload support via st.file_uploader(accept_multiple_files=True) rendering dynamic per-file tabs.
? Real-time confidence threshold slider (5% to 95%) routing low-confidence predictions to 'Uncategorized'.
? Interactive st.data_editor table enabling manual category overrides with SelectboxColumn before export.
? Interactive Plotly analytics: Expense breakdown donut chart and daily cash flow inflow/outflow bars.

2. Hybrid Ingestion & Per-Page Routing Pipeline

? inspect_pdf_pages: Inspects every page independently using PyMuPDF to classify as text, scanned, or empty.
? Per-Page Hybrid Routing: Text pages route to high-speed pdfplumber table extraction; scanned pages route to 300 DPI rendering + OpenCV deskewing + Tesseract OCR.
? Unified Document Merger: Assembles chronological transactions and parses Account Details (Name, A/C #, IFSC, Bank) from combined text.

3. Resilient OCR & Deskewing Subsystem

? OpenCV Deskewing: Calculates horizontal projection profile variance across angles [-5°, +5°] and applies affine rotation to straighten tilted scans.
? Image Preprocessing: Grayscale conversion, contrast adjustment, and sharpening preserve fine numerical punctuation.
? Right-to-Left Line Parser: Handles OCR placeholders (_, ?, ?, -, ~), recognizes word boundaries (\bCR\b vs 'SUBSCRIPTION'), and tracks running balance differentials.

4. Dual Non-LLM Classification Engines

? Engine A (Rule-Based): 16 financial categories using strict \b...\b word boundaries. Two-tier evaluation ensures specific merchant keywords strictly take priority over channel tokens (UPI, NEFT, IMPS, POS). Includes direction awareness (refunds -> Income, debit interest -> Loans).
? Engine B (Machine Learning): FeatureUnion combining word n-grams (1,2) and char_wb n-grams (3,5) with Logistic Regression. Trained on 403 realistic banking descriptions. Evaluated via 5-Fold Stratified Cross-Validation: 91.1% Accuracy, 90.9% F1-score.

5. Balance Audit, Reconciliation & Multi-Sheet Export

? Row-by-Row Mathematical Validation: Verifies Balance_i = Balance_{i-1} - Debit_i + Credit_i with exact variance logging.
? Non-Circular Reconciliation: Verifies Closing Balance = Opening Balance - Total Debits + Total Credits.
? Multi-Sheet Excel Export: Sheet 1 contains Account Metadata & Audit Reconciliation; Sheet 2 contains styled transactions with currency formatting.
? Clean CSV Export: RFC-4180 compliant UTF-8 CSV with all validated transaction details and confidence scores.